General Bite #9: Modulation and Bandwidth

AM/FM/SSB carrier modulation, modulation envelope, FSK/PSK/QPSK, mixer products, bandwidth math, intermod and image response.

Groups G8A + G8B · 27 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

Part 1: Modulation Methods (G8A)

The Three Classic Modulations

Type	What varies	Q
AM (amplitude modulation)	**Varies the instantaneous power level of the RF signal** (G8A05). Information is in the envelope.	G8A05
FM (frequency modulation)	**Changes the instantaneous frequency of an RF wave** (G8A03).	G8A03
PM (phase modulation)	**Changes the phase angle of an RF signal** (G8A02). A reactance modulator on an RF amplifier stage produces **phase modulation** (G8A04).	G8A02, G8A04

Digital Modulation Snapshots

Question	Answer	Q
How direct binary FSK is generated	**By changing an oscillator's frequency directly with a digital control signal** (G8A01).	G8A01
QPSK definition	**Digital data transmitted using 0/90/180/270° phase shifts to represent pairs of bits** (G8A12). Four states, two bits per symbol.	G8A12
QPSK31 characteristic	Sideband-sensitive + encoding provides error correction + bandwidth BPSK31 – **all of the above** (G8A06).	G8A06
FT8 modulation type	**8-tone frequency shift keying** (G8A09).	G8A09

AM Envelope, Overmodulation, Flat-Topping

Question	Answer	Q
AM modulation envelope	**The waveform created by connecting the peak values of the modulated signal** (G8A11).	G8A11
Effect of overmodulation	**Excessive bandwidth** (G8A08). Sprays splatter into adjacent frequencies.	G8A08
"Flat-topping" on an AM phone signal	**Signal distortion caused by excessive drive or speech levels** (G8A10). The peaks get clipped.	G8A10
Narrowest phone bandwidth	**Single sideband** (G8A07).	G8A07

Link Budget And Link Margin

Link budget = transmit power + antenna gains − system losses as seen at the receiver (G8A13).
Link margin = received power level − minimum required signal level at the receiver input (G8A14). The dB cushion above the threshold of intelligible reception.

Part 2: Frequency Conversion And Bandwidth (G8B)

Mixers And Heterodyning

Question	Answer	Q
Which mixer input is tuned to convert different RF signals to IF?	**Local oscillator** (G8B01).	G8B01
Output of a mixer = …	**The sum and difference** of the two input frequencies (G8B11).	G8B11
Term for mixing two RF signals	**Heterodyning** (G8B03).	G8B03
Image response	**Interference from a signal at twice the IF frequency from the desired signal** (G8B02). Two frequencies map to the same IF – the wanted and the image.	G8B02
Stage that generates a harmonic to reach a higher operating frequency	**Multiplier** (G8B04). Common in VHF FM transmitter chains.	G8B04

Intermodulation

What process combines two signals in a non-linear circuit to produce unwanted spurious outputs? Intermodulation (G8B12).
Which intermod products are closest to the original signal frequencies? Odd-order (G8B05). They land in-band and are the most dangerous.
Example odd-order product of F1 and F2: 2F1−F2 (G8B13). Order = sum of the absolute coefficients. 2F1−F2 is 3rd order (odd). 3F1−F2 is 4th order, 5F1−3F2 is 8th order – both even.

Bandwidth Math

FM bandwidth (Carson's rule) ≈ 2 × (deviation + max audio frequency).
G8B06 worked: 5 kHz deviation + 3 kHz audio: BW = 2 × (5 + 3) = **16 kHz** (G8B06).
G8B07 worked: a VHF transmitter at 146.52 MHz with 5 kHz deviation uses an oscillator at 12.21 MHz multiplied 12× to reach 146.52 MHz. Deviation multiplies by the same factor. So the oscillator deviation = 5000 Hz ÷ 12 ≈ **416.7 Hz** (G8B07).

Why Bandwidth Matching Matters

Question	Answer	Q
Why know your mode's duty cycle?	**Some modes have high duty cycles that could exceed the transmitter's average power rating** (G8B08).	G8B08
Why match receiver bandwidth to the mode's bandwidth?	**Results in the best signal-to-noise ratio** (G8B09).	G8B09
Symbol rate vs bandwidth	**Higher symbol rates require wider bandwidth** (G8B10).	G8B10

Mental Cheat Sheet

- AM varies **amplitude/power**; FM varies **frequency**; PM varies **phase**.
- Reactance modulator on RF amp = **PM**.
- FT8 uses **8-tone FSK**.
- QPSK = 4-phase, 2 bits/symbol.
- SSB is the **narrowest** phone mode.
- AM envelope = curve through peak amplitudes.
- Overmodulation → **excessive bandwidth**.
- Flat-topping = **excessive drive/speech**.
- Link budget = TX + gains − losses at RX.
- Link margin = received level − minimum required.
- Mixer: tune **local oscillator** to bring signal to IF.
- Mixer output = **sum + difference** of inputs.
- Mixing two RF signals = **heterodyning**.
- Image response = signal **2 × IF away** from desired.
- Multiplier stage = generates **harmonics**.
- Intermod = non-linear mixing → spurious.
- **Odd-order** products land closest to the originals.
- FM BW (Carson) = $2(\Delta f + f_m)$. 5 + 3 → **16 kHz**.
- Multiplier multiplies deviation; oscillator deviation = output deviation / multiplier factor.
- High duty cycle → watch **average power** rating.
- Match RX BW to mode for best **SNR**.
- Higher symbol rate → wider bandwidth.

Common Traps

- Reactance modulator on the RF amp gives **PM**, not FM (FM is reactance modulator on the **oscillator**).
- FT8 is **8-tone FSK**, not direct-sequence spread spectrum.
- SSB is **narrowest**, not FM or VSB.
- Overmodulation → **excessive bandwidth**, not "insufficient audio."
- Flat-topping is **clipping from excessive drive**, not "ALC properly adjusted."
- Link budget uses **all** gains/losses; link margin is the **cushion above MDS**.
- Image response is at **2 × IF** from the desired signal, not 2× the signal frequency.
- Mixer output is **sum and difference**, not the **product**.
- Intermod = **non-linear circuits**, not "positive feedback."
- Odd-order intermod products are dangerous because they land **in-band**.
- Higher symbol rate → **wider** bandwidth, not narrower.
- High duty cycle stresses the **average power** rating, not PEP.

Bite 9 Question Bank (G8A + G8B)

G8A01. How is direct binary FSK modulation generated?
A. By keying an FM transmitter with a sub-audible tone.
B. By changing an oscillator's frequency directly with a digital control signal.
C. By using a transceiver's computer data interface protocol to change frequencies.
D. By reconfiguring the CW keying input to act as a tone generator.

G8A02. What is the name of the process that changes the phase angle of an RF signal to convey information?
A. Phase convolution.
B. Phase modulation.
C. Phase transformation.
D. Phase inversion.

G8A03. What is the name of the process that changes the instantaneous frequency of an RF wave to convey information?
A. Frequency convolution.
B. Frequency transformation.
C. Frequency conversion.
D. Frequency modulation.

G8A04. What emission is produced by a reactance modulator connected to a transmitter RF amplifier stage?
A. Multiplex modulation.
B. Phase modulation.
C. Amplitude modulation.
D. Pulse modulation.

G8A05. What type of modulation varies the instantaneous power level of the RF signal?
A. Power modulation.
B. Phase modulation.
C. Frequency modulation.
D. Amplitude modulation.

G8A06. Which of the following is characteristic of QPSK31?
A. It is sideband sensitive.
B. Its encoding provides error correction.
C. Its bandwidth is approximately the same as BPSK31.
D. All these choices are correct.

G8A07. Which of the following phone emissions uses the narrowest bandwidth?
A. Single sideband.
B. Vestigial sideband.
C. Phase modulation.
D. Frequency modulation.

G8A08. Which of the following is an effect of overmodulation?
A. Insufficient audio.
B. Insufficient bandwidth.
C. Frequency drift.
D. Excessive bandwidth.

G8A09. What type of modulation is used by FT8?
A. 8-tone frequency shift keying.
B. Vestigial sideband.
C. Amplitude compressed AM.
D. 8-bit direct sequence spread spectrum.

G8A10. What is meant by the term “flat-topping,” when referring to an amplitude-modulated phone signal?
A. Signal distortion caused by insufficient collector current.
B. The transmitter’s automatic level control (ALC) is properly adjusted.
C. Signal distortion caused by excessive drive or speech levels.
D. The transmitter’s carrier is properly suppressed.

G8A11. What is the modulation envelope of an AM signal?
A. The waveform created by connecting the peak values of the modulated signal.
B. The carrier frequency that contains the signal.
C. Spurious signals that envelop nearby frequencies.
D. The bandwidth of the modulated signal.

G8A12. What is QPSK modulation?
A. Modulation using quasi-parallel to serial conversion to reduce bandwidth.
B. Modulation using quadra-pole sideband keying to generate spread spectrum signals.
C. Modulation using Fast Fourier Transforms to generate frequencies at the first, second, third, and fourth harmonics of the carrier frequency to improve noise immunity.
D. Modulation in which digital data is transmitted using 0-, 90-, 180- and 270-degrees phase shift to represent pairs of bits.

G8A13. What is a link budget?
A. The financial costs associated with operating a radio link.
B. The sum of antenna gains minus system losses.
C. The sum of transmit power and antenna gains minus system losses as seen at the receiver.
D. The difference between transmit power and receiver sensitivity.

G8A14. What is link margin?
A. The opposite of fade margin.
B. The difference between received power level and minimum required signal level at the input to the receiver.
C. Transmit power minus receiver sensitivity.
D. Receiver sensitivity plus 3 dB.

G8B01. Which mixer input is varied or tuned to convert signals of different frequencies to an intermediate frequency (IF)?
A. Image frequency.
B. Local oscillator.
C. RF input.
D. Beat frequency oscillator.

G8B02. What is the term for interference from a signal at twice the IF frequency from the desired signal?
A. Quadrature response.
B. Image response.
C. Mixer interference.
D. Intermediate interference.

G8B03. What is another term for the mixing of two RF signals?
A. Heterodyning.
B. Synthesizing.
C. Frequency inversion.
D. Phase inversion.

G8B04. What is the stage in a VHF FM transmitter that generates a harmonic of a lower frequency signal to reach the desired operating frequency?
A. Mixer.
B. Reactance modulator.
C. Balanced converter.
D. Multiplier.

G8B05. Which intermodulation products are closest to the original signal frequencies?
A. Second harmonics.
B. Even-order.
C. Odd-order.
D. Intercept point.

G8B06. What is the total bandwidth of an FM phone transmission having 5 kHz deviation and 3 kHz modulating frequency?
A. 3 kHz.
B. 5 kHz.
C. 8 kHz.
D. 16 kHz.

G8B07. What is the frequency deviation for a 12.21 MHz reactance modulated oscillator in a 5 kHz deviation, 146.52 MHz FM phone transmitter?
A. 101.75 Hz.
B. 416.7 Hz.
C. 5 kHz.
D. 60 kHz.

G8B08. Why is it important to know the duty cycle of the mode you are using when transmitting?
A. To aid in tuning your transmitter.
B. Some modes have high duty cycles that could exceed the transmitter’s average power rating.
C. To allow time for the other station to break in during a transmission.
D. To prevent overmodulation.

G8B09. Why is it good to match receiver bandwidth to the bandwidth of the operating mode?
A. It is required by FCC rules.
B. It minimizes power consumption in the receiver.
C. It improves impedance matching of the antenna.
D. It results in the best signal-to-noise ratio.

G8B10. What is the relationship between transmitted symbol rate and bandwidth?
A. Symbol rate and bandwidth are not related.
B. Higher symbol rates require wider bandwidth.
C. Lower symbol rates require wider bandwidth.
D. Bandwidth is half the symbol rate.

G8B11. What combination of a mixer’s Local Oscillator (LO) and RF input frequencies is found in the output?
A. The ratio.
B. The average.
C. The sum and difference.
D. The arithmetic product.

G8B12. What process combines two signals in a non-linear circuit to produce unwanted spurious outputs?
A. Intermodulation.
B. Heterodyning.
C. Detection.
D. Rolloff.

G8B13. Which of the following is an odd-order intermodulation product of frequencies F1 and F2?
A. 5F1-3F2.
B. 3F1-F2.
C. 2F1-F2.
D. All these choices are correct.

Answer Key

G8A01: B	G8A02: B	G8A03: D	G8A04: B	G8A05: D
G8A06: D	G8A07: A	G8A08: D	G8A09: A	G8A10: C
G8A11: A	G8A12: D	G8A13: C	G8A14: B	G8B01: B
G8B02: B	G8B03: A	G8B04: D	G8B05: C	G8B06: D
G8B07: B	G8B08: B	G8B09: D	G8B10: B	G8B11: C
G8B12: A	G8B13: C			